A_GMX in brief

the algorithm as implemented, and the comparison with WADERs

comm_scripts clustering study

code of 26 October 2026

Code: `clustering_tests/adaptive.py` (`adaptive_cluster(..., engine="gmx")`), `clustering_tests/gmx.py` (`general_cluster`), `clustering_tests/agent_methods.py` (`background_map`), `clustering_tests/fe_candidates.py`; run with `run_AGMX` or `run_WADERs --clusterer agmx`. The full derivations, the history and every test are in `docs/aGMX/aGMX_clustering.pdf`.

1 Input, output, notation

Input: a calibrated image Q_p in electrons (panaSKImg's `*_EXT<n>_waders.fits.bz2`, converted with the module's $g(n)$ curve) and its active-pixel mask. Output: one record per cluster (energy, position, width, class, quality flags, pixels with the charge taken from each) in the WADERs ROOT format, and for electron-counting CCDs a few-electron candidate catalogue. Energy = charge $\times$ 3.74 eV/e^-.

Pixel model: $Q_p = o_p + n_p + \epsilon_p + D_p$ with offset o_p, dark charge $n_p \sim \text{Poisson}(\lambda_p)$, read noise $\epsilon_p \sim \mathcal{N}(0, s_p^2)$ and deposit charge D_p. Derived maps: $R_p = Q_p - o_p - \lambda_p$ (residual), $V_p = \lambda_p + s_p^2$ (variance of an empty pixel), $R_p^+ = \max(R_p, 0)$.

2 The algorithm

0. Background

- **Row fit.** Per row, the pixels away from clusters are fitted with the comb $P(Q) = \sum_n \text{Pois}(n;\lambda)\,\mathcal{N}(Q; o+n, s^2)$. Fast path: the comb phase $z = \langle e^{2\pi i Q}\rangle$ gives $o = \arg z/2\pi$ and $s = \sqrt{-\ln|z|/2\pi^2}$, and λ comes from the rounded counts. A row counts as resolved only if $|z| > \max(e^{-2\pi^2\,0.45^2}, 4/\sqrt{N})$, i.e. above the noise floor $\sqrt{\pi/4N}$ of N unresolved pixels.
- **Unresolved rows** (comb washed out, e.g. by a charge sheet) take o, λ, s from the nearest resolved rows when at least max(10, 5 %) of the rows are resolved. Otherwise a pooled comb likelihood is fitted over groups of rows.
- **Rounding correction.** On fast-path rows, noise above $+0.5\,e^-$ is counted as an electron: $\lambda \leftarrow (\lambda - p)/(1-2p)$ with $p = \Phi^c(0.5/s)$.
- λ is smoothed along the rows (local-linear) and in column blocks, with floor $\lambda \geq 10^{-6}$. Cluster pixels are masked by a Poisson-tail test and the fit is iterated.
- **Hot columns.** Per column, a clipped mean of $Q - o - \lambda$ (one 5σ clip around the column median) is compared with the median of the ± 10 neighbouring columns. A column whose excess d is $> 5\times$ its error and $> 0.05\,e^-$ gets $\lambda \leftarrow \lambda + d$, so the mean, the variance and the Poisson tests all include it. Nothing is flagged if more than 10 % of the columns would be (a charge sheet, not hot columns).
- **Read noise** σ_{read}: 1.4826$\times$ the MAD of the pixels with $|Q| < 0.5\,e^-$. If that exceeds $0.3\,e^-$, it is max(itself, MAD of all active pixels).

A. Switch

The image is "unmodelled" when $\big(s \geq 0.5\,e^-$ and $s \geq 3\sigma_{\text{read}}\big)$ or $\sigma_{\text{read}} \geq 1\,e^-$, where s is the median of the fitted s_p. It is then clustered on Q by hysteresis with σ_z, the zero-peak MAD:

- pixels $> \max(10\sigma_z, 4.5\,e^-)$ that touch (8-connected) form a cluster;

- it is kept if a pixel is $> \max(30\sigma_z,\, 9.5\,e^-)$;
- $E = \sum Q_p$.

Otherwise GMX (steps 1–8) runs on the whole image. Every cluster also reports the event occupancy of its 31×31-px neighbourhood ("crowded" if ≥ 0.3); this is information only.

1. Seeds

- **Matched filter** at widths $w \in \{0.5, 1, 1.5, 2.5\}$ px, with h_w the pixel-integrated unit Gaussian: $T_w(c) = \sum_p h_w(p-c)R_p \big/ \sqrt{\sum_p h_w(p-c)^2 V_p}$.
- **Seeds** are local maxima (5×5) with $T_w > 5$, taken finest scale first. A coarser maximum is kept only if no finer seed lies within $\max(2, 1.5w)$ px.
- **Poisson check** (each seed): on the 3×3 or the 5×5 box, with $S = \sum(Q-o)$, $\Lambda = \sum\lambda$ and $\sigma_\mathrm{b} = \sqrt{\sum s^2}$, require $P\big(\mathrm{Pois}(\Lambda) \geq \lfloor S - 2\sigma_\mathrm{b} + 0.5 \rfloor\big) < \Phi^c(5)$. On counting CCDs λ is the detection dark charge (step 7).
- **Initial cloud:** centre = the seed pixel, $\Sigma = w^2\mathbf{1}$, $A_0 = \max\big(R_\mathrm{seed}\, 2\pi(w^2 + \frac{1}{12}),\, 1\big)$.

2. Mixture fit (EM)

- **Model:** $M_p = \sum_k g_k(p)$. For a cloud with correlation $|\rho| < 0.5$ (every X-ray), $g_k(p)$ is the exact pixel integral
$$g_k(p) = A_k\big[\Phi(\tfrac{i+\frac{1}{2}-\mu_y}{\sigma_y}) - \Phi(\tfrac{i-\frac{1}{2}-\mu_y}{\sigma_y})\big]\big[\Phi(\tfrac{j+\frac{1}{2}-\mu_x}{\sigma_x}) - \Phi(\tfrac{j-\frac{1}{2}-\mu_x}{\sigma_x})\big].$$
Strongly correlated clouds (pieces of tilted tracks) use $A_k\,\mathcal{N}(\mathbf{x}_p; \boldsymbol{\mu}_k, \Sigma_k + \frac{1}{12}\mathbf{1})$. A cloud is evaluated within its 4σ ellipse (radius 2–16 px).
- **E-step:** $r_{pk} = g_k(p)/(M_p + \varepsilon_p)$, with the "unexplained" level $\varepsilon_p = \sqrt{V_p}$, so noise pixels far from every cloud belong to none.
- **M-step:** with $w_{pk} = r_{pk}R_p^+$, $A_k = \sum_p w_{pk}$, and the weighted mean and covariance of the pixel positions. For $|\rho| < 0.5$ clouds the per-axis mean and variance are inverted through tables of the pixelised Gaussian (they depend on σ and the sub-pixel position). Otherwise $\frac{1}{12}$ is subtracted from the variances.
- **Safeguards:** variances in $[0.02, 25]\,\mathrm{px}^2$, $|\sigma_{xy}| \leq 0.95\sqrt{\sigma_{yy}\sigma_{xx}}$, centre step ≤ 1.5 px per iteration; clouds with $A < 10^{-3}\,e^-$ are removed. 30 iterations, fixed (running to convergence gave no consistent gain). Compiled with numba; `AGMX_NUMBA=0` gives the identical numpy version.

3. Residual pass and pruning

- **Residual pass.** The matched filter is run on $R - M$ with variance $V + M$ at $w = 0.5, 1$. Local maxima (3×3) with $T > 5$, passing the Poisson check (counting CCDs) and more than 1 px from every cloud, become new clouds. Then 15 EM iterations.
- **Pruning.** Clouds with $\mathrm{S/N}_k = A_k\sqrt{\sum_p \pi_k(p)^2/(V_p + M_p - g_k(p))} < 4$ are dropped, where $\pi_k = g_k/A_k$. Then 5 EM iterations.

4. Grouping clouds into deposits

Every pair closer than $3.5(\bar\sigma_i + \bar\sigma_j) + 1$ px is tested ($\bar\sigma$: geometric-mean width, ≤ 16 px).

- **Ridge test:** $\rho_{ij} = \min_t f(t)/\min(f(0), f(1)) > 0.5$, where f is the two-cloud density along the line joining the centres (17 points).
- **Shape test**, one of: width ratio < 1.6; a track (one cloud has axis ratio > 1.4 and its long axis is within $|\cos| > 0.8$ of the joining line); or the narrower cloud is not a local maximum of f (a shoulder).
- **Fragments.** A cloud below $50\,e^-$ *and* below 10 % of its brightest merge partner links only to that partner, so it cannot join an X-ray to a neighbouring large deposit.

Clusters are the connected components of the links.

5. Island test (tracks versus several X-rays)

Islands are 8-connected regions above the Poisson-tail threshold of the dark charge ($\alpha = 0.01$), of at most 4000 px. An island holding $K \geq 2$ clusters is refitted on its pixels (+2 rings) under two hypotheses:

- K round Gaussians (4 parameters each);
- the free clouds already fitted (6 parameters each, n_c clouds).

They are compared by

$$\Delta\mathrm{BIC} = \chi^2_{\mathrm{round}} + 4K \ln n - \chi^2_{\mathrm{free}} - 6n_c \ln n, \qquad \chi^2 = \sum_p \frac{(R_p - M_p)^2}{V_p + R_p^+ + (0.05R_p)^2}.$$

If $\Delta\mathrm{BIC} > 100$ the whole island becomes one deposit (30 with `AGMX_PIX_EXACT=0`).

6. Energy, width and flags

- **Energy by sharing:** $E_c = \sum_p \tilde{r}_{pc} R_p$, with $\tilde{r}_{pc} = \sum_{k \in c} g_k(p)/M_p$. The measured charge is split, never counted twice.
- **Satellites.** A cluster with $E < 1\,\%$ of a neighbour's *and* $E < 1\,\mathrm{keV}$ is merged into that neighbour if its centre lies in the neighbour's domain or within 6σ (Mahalanobis) of one of its clouds (within 20 px).
- **Growth.** Pixels above the $\alpha = 0.01$ Poisson tail, within 3 px of a covered pixel, connected to it and covered by no cloud (charge-transfer trails), are added to the nearest cluster.
- **Width:** $\sigma_{xy}^2 = \frac{1}{2}(\mathrm{var}_y + \mathrm{var}_x) - \frac{1}{12}$, the second moments of the cluster's shared charge. A single cloud whose EM width is < 0.4 px reports the EM width, because the moment is noise-dominated there.
- **Flags**, which never change the energy:
  - blend $= \sum_p M_p^{(c)} M_p^{(\neg c)} / \sum_p (M_p^{(c)})^2$ (other clusters' charge under this one's profile);
  - ped: the 25th percentile of R in a ring 5–8 px around the centre (a charge sheet);
  - peakfrac: the largest cloud's share of the fitted charge;
  - χ^2: $\sum (R-M)^2/(V+M)$ per pixel over the pixels the cluster owns ($\tilde{r} > 0.5$);
  - sig: $E_c / \sqrt{\sum \tilde{r}^2 V}$;
  - edge: within 2 px of the mask;
  - saturated: a pixel at an ADC-limit plateau.

7. Few-electron events (median $s < 0.35\,e^-$)

- **Detection dark charge** λ_{det}: λ implied by the rate of *isolated* single-electron pixels (no counted pixel within 2 px; smoothed over ± 30 rows; rounding miscounts included), never below the fitted map, plus the hot-column excess.
- **Counting energies.** A cluster below $100\,e^-$ gets $E = \sum_{\tilde{r}>0.5} \big(\mathrm{round}(Q - o) - \lambda\big)$; the analog value is kept as `E_fit`.
- **Counting finder.** An 8-connected group of ≤ 9 pixels with $\mathrm{round}(Q-o) \geq 1$, not covered by a cluster, has statistic $T = -\sum_i \ln P(n_i \,|\, \lambda_{\mathrm{det},i}, s_i) - \ln N_{\mathrm{shapes}}(m)$, with $P(n|\lambda, s) = \sum_k \mathrm{Pois}(k; \lambda)\big[\Phi(\frac{n+\frac{1}{2}-k}{s}) - \Phi(\frac{n-\frac{1}{2}-k}{s})\big]$. The group is kept when T exceeds the value exceeded by 0.08 groups per Mpix in 10–50 blank images simulated from the image's own $\lambda_{\mathrm{det}}, s$ (fixed seed). The analytic bound $\ln(N_{\mathrm{pix}}/F)$ is used only if the blanks have too few groups. Class "low-E".

8. Classes

- **low-E:** $E < 0.5\,\mathrm{keV}$.
- **x-ray / x-ray diffuse:** one round cloud or extent < 6 px; diffuse if $\sigma_{xy} \geq 1$.
- **alpha:** $> 1\,\mathrm{MeV}$.
- **artifact-row / -column:** thin, straight and aligned with a row or column.
- **muon:** straight skeleton (≥ 10 px, scatter of the cloud centres < 1 px and $< 6\,\%$ of the length) *and* $\geq 2.5\,\mathrm{keV}$ per projected pixel; a straight track below that rate is "extended".
- **worm:** everything else extended.

9. After A_GMX: the few-electron candidate catalogue

For counting CCDs, written to `fe_candidates/*_fe_candidates.csv` and `.json` (off with `--no-fe-candidates`):

- **Candidates:** every group of 2–20 counted electrons (≤ 9 px), with T, $n_{\rm exp}$ (expected dark-charge groups per image at least as unlikely; simulated blanks of the image's own occupancy-based λ) and $p_{\rm image} = 1 - e^{-n_{\rm exp}}$.
- **Mask flags:** halo ($5\sqrt{E/\mathrm{keV}}$ px, 5–60) around clusters, row/column trails (50 px, $E > 1$ keV), hot columns, edge (5 px), overlap with a cluster.
- **Per image:** observed and expected clean counts per class (charge, pixels).

Nothing is removed; it is the input for a rare-event analysis.

Recommended selection for X-ray lines: $\sigma_{xy} < 0.7$ (Am-241) or < 1 (front-side Fe-55), blend < 0.02, ped $< 10\,e^-$/pix, not edge.

Switches (environment): `AGMX_PIX_EXACT=0` (Gaussian-at-centre model, BIC margin 30, no fragment rule), `AGMX_PHYS_FIXES=0` (behaviour before the 25–26 October audit), `AGMX_NUMBA=0`, `AGMX_FE_CANDIDATES=0`.

3 Comparison with WADERs

WADERs: panaSKImg's ClusterFinder, i.e. hysteresis with pixel $> 10\sigma$ and seed $> 30\sigma$ of the pedestal, $E = \sum$ core pixels. The tests below use the same injected objects and the same images for both methods. **Code versions:** the crowded-CCD fit is from the current code (26 October). The wide, Fe-55-width, track and few-electron tests were run on the code of 25 October, before the physics audit and the exact pixel integrals. On a subset (80 image-CCDs) the current code changes front-side Kα by -0.2 eV (median) and resolves more pairs (Fe-55 0.907→0.941, Am-241 0.607→0.655). A full rerun of those tests is pending.

	WADERs	A_GMX	code
Crowded Am-241 CCD (AMtest EXT4, 29 images), constrained 10–30 keV fit (Lépy line ratios)			
energy scale k (1 = right)	0.9840	**1.0006**	26 Oct
Np Lβ / Lγ / γ 26.3 / Ag K significance [σ]	15.5 / 16.0 / 14.8 / 12.9	**22.4 / 20.6 / 21.1 / 18.3**	26 Oct
combined Z (Np L + γ 26.3)	27.1	**37.4**	26 Oct
deviance / ndf	**1.74**	2.59	26 Oct
Fe-55 (DM09/DM10, 220 injected clusters per width and CCD; 8 CCDs)			
energy of injected clusters, width 0.3 → 1.8 px (median)	0.997 → 0.921	**1.000 → 1.000**	25 Oct
worst CCD at width 1.8 px	0.737	**0.994**	25 Oct
real front-side Kα / 5.899 keV: mean, spread over 8 CCDs	0.9928, 1.32 %	**0.9969, 0.23 %**	25 Oct
two diffuse (σ 1.4 px) clusters 5 px apart resolved	0.00	**0.93**	25 Oct
Wide test: 878 image-CCDs, 16 data sets, injected objects			
Fe-55 / muon / worm / 4-px pair success	0.86 / 0.59 / 0.74 / 0.11	**0.93 / 0.64 / 0.79 / 0.88**	25 Oct
Fe-55 energy resolution (rms of $E/E_{\rm true}$)	0.52 %	**0.19 %**	25 Oct
dark charge $\geq 1\,e^-$/pix: Fe-55 / worm / pair	0.55 / 0.45 / 0.07	**0.87 / 0.74 / 0.79**	25 Oct
Injected objects in real Am-241 images (EXT1)			
alpha / low-E / 3-px pair / 4-px pair	0.90 / 0.47 / 0.00 / 0.21	**0.94 / 0.64 / 0.75 / 0.93**	25 Oct
muon / worm / hot column	0.90 / 0.93 / **0.76**	0.93 / 0.93 / 0.74	25 Oct
Few-electron events (simulated, 0.15 e^- noise)			
3-e^- found at $10^{-3}\,e^-$/pix	0.00	**0.85** (fakes 0.07 per 1.7 Mpix)	26 Oct
3-e^- found at $10^{-4}\,e^-$/pix	not run	**0.96** (fakes 0.30 per 1.7 Mpix)	26 Oct
run time per image-CCD (median)	**0.01 s**	0.7 s (Fe-55 0.9 s, Am-241 62 s)	25/26 Oct

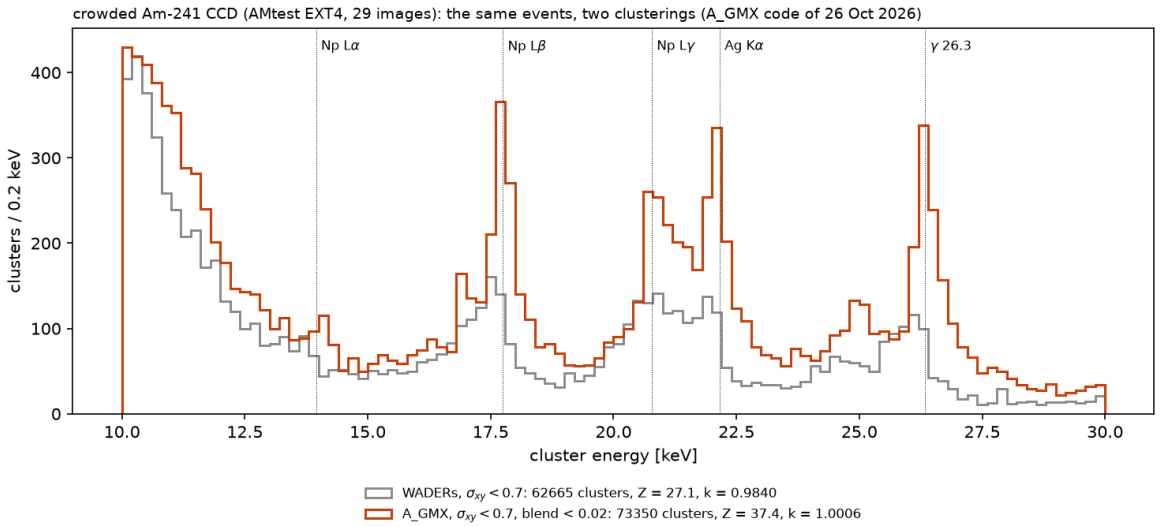


Figure 1: The crowded CCD, current code: the same events clustered by WADERs and by A_GMX (blend < 0.02). A_GMX's lines are higher above the continuum and at the right energy ($k = 1.0006$ vs 0.984). Not yet understood: its spectrum between the lines fits worse (deviance/ndf 2.59 vs 1.74).

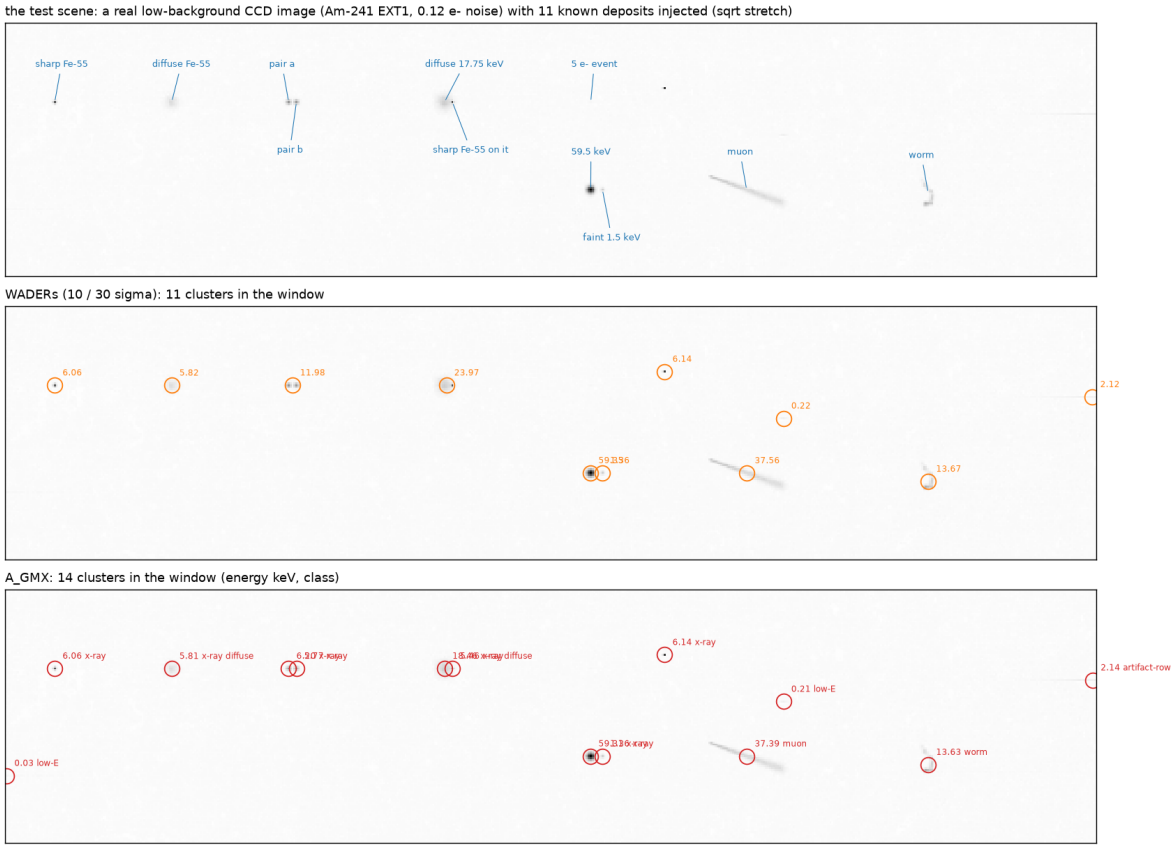


Figure 2: A real Am-241 image (EXT1, $0.12\,e^-$ noise) with eleven known deposits injected (top). WADERs (middle) merges a 4-px pair into one 12-keV cluster, and a sharp X-ray on a diffuse one into one 24-keV cluster; A_GMX (bottom) separates both. Code of 25 October.

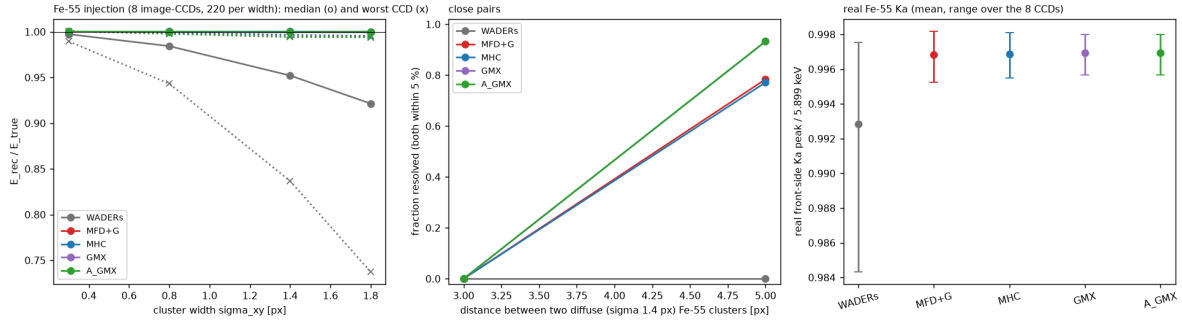


Figure 3: Fe-55. Left: median (circles) and worst-CCD (crosses) $E_{\mathrm{rec}}/E_{\mathrm{true}}$ against cluster width; WADERs' thresholds cut the tails of diffuse clusters. Middle: diffuse pairs resolved. Right: real front-side Kα over the 8 CCDs. Code of 25 October.

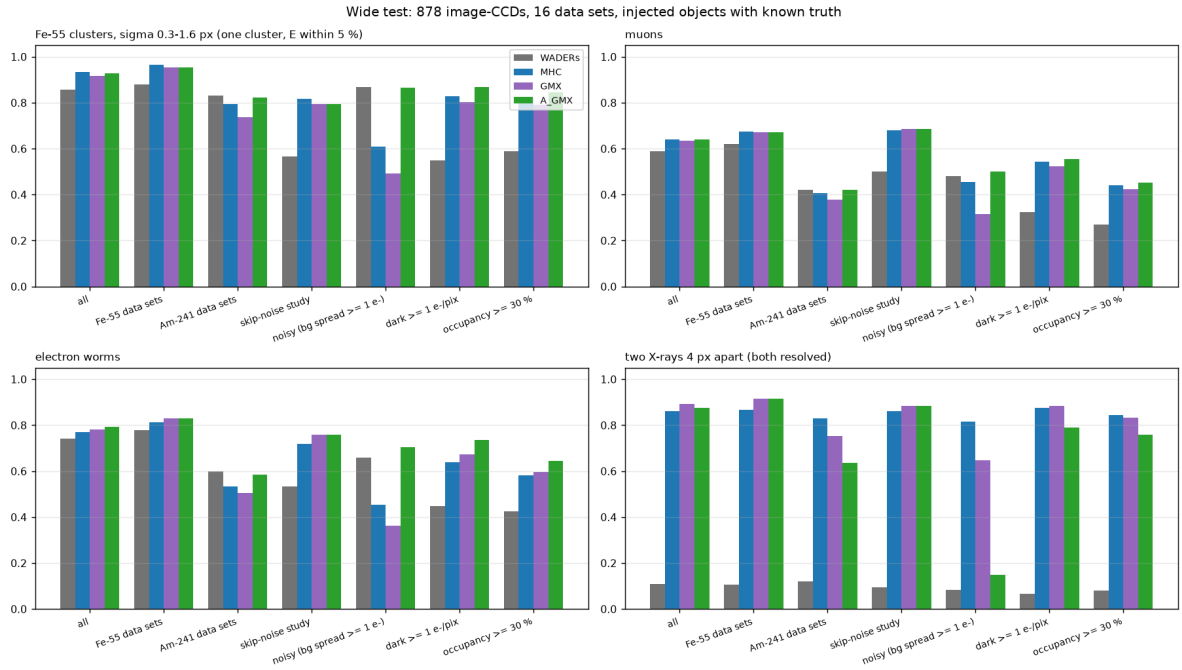


Figure 4: Wide test (878 image-CCDs): success per object type and image condition. Code of 25 October.

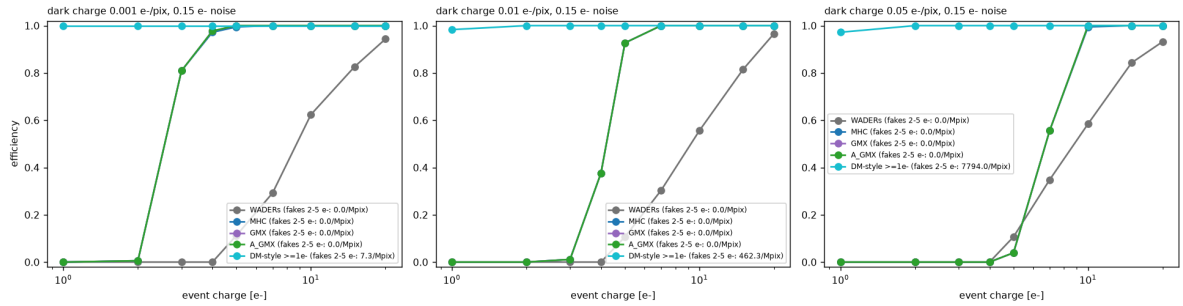


Figure 5: Few-electron efficiency against event charge at three dark charges (code of 25 October). WADERs needs $\gtrsim 10\,e^-$. The current code's detection background and calibrated threshold raise A_GMX's 3-e^- efficiency at the lowest dark charges (table above).

Where WADERs is still the better choice:

- speed (about 70× faster);
- the noisy "unmodelled" images (read noise $\geq 1\,e^-$), where A_GMX is WADERs-like hysteresis anyway;
- hot-column artifacts (0.76 vs 0.74);

- the smoothness of the crowded-CCD spectrum between the lines (deviance/ndf).

Open items:

- the higher deviance/ndf on the crowded CCD;
- thresholds for noisy images (pixel $\geq 4.5\,e^-$ is a stop-gap);
- the background fixes raise the fitted dark charge on the crowded CCD ($0.093 \rightarrow 0.157\,e^-$/pix median), which shifts the 59.5-keV peak by about $+0.1$ keV;
- recalibrating the χ^2 flag for the exact model;
- rerunning the wide, Fe-55 and track tests on the current code.